Numbers, Percents & Ratios

ANSWER KEY

Properties of integers and number classification

1 Classify the number $-\frac{7}{2}$ as an integer, rational number, or irrational number.
$-\frac{7}{2} = -3.5$ is a rational number (it can be expressed as a ratio of two integers), but it is NOT an integer, since it is not a whole number.

2 Find the greatest common factor (GCF) and least common multiple (LCM) of 18 and 24.
$18 = 2 \times 3^2$, $24 = 2^3 \times 3$. GCF $= 2 \times 3 = 6$. LCM $= 2^3 \times 3^2 = 8 \times 9 = 72$.

Divisibility and prime factorization

3 Find the prime factorization of 360.
$360 = 2^3 \times 3^2 \times 5$.

4 Determine whether 91 is prime, and justify your answer.
$91 = 7 \times 13$, so 91 is not prime — it has factors other than 1 and itself.

Percent increase and decrease

5 A price increases from 80 dollars to 92 dollars. Find the percent increase.
Increase: $92 - 80 = 12$. Percent increase: $\frac{12}{80} \times 100\% = 15\%$.

6 A population of 5000 decreases by 8%. Find the new population.
Decrease: $5000 \times 0.08 = 400$. New population: $5000 - 400 = 4600$.

Percent of a percent (successive percent changes)

7 A jacket is marked down 30%, then an additional 10% is taken off the sale price. If the original price was 100 dollars, find the final price.
After the first discount: $100(0.70) = 70$ dollars. After the second discount: $70(0.90) = 63$ dollars.

Ratios and proportional reasoning

8 The ratio of boys to girls in a class is 3:5. If there are 40 students total, find the number of boys.
Total ratio parts: $3 + 5 = 8$. Each part represents $\frac{40}{8} = 5$ students. Boys: $3 \times 5 = 15$.

9 A recipe uses a ratio of 2 cups flour to 3 cups sugar. If a baker uses 15 cups of sugar, how much flour is needed?

Set up a proportion: $\frac{2}{3} = \frac{x}{15}$. Cross-multiply: $3x = 30$, so $x = 10$ cups of flour.

Scientific notation

10 Write 0.000452 in scientific notation.

4.52×10^{-4}.

11 Evaluate $(3 \times 10^{4}) \times (2 \times 10^{-2})$, giving your answer in scientific notation.

$(3 \times 2) \times 10^{4+(-2)} = 6 \times 10^{2}$.

Absolute value and number line reasoning

12 Find all values of x satisfying $|x - 4| = 9$.

Either $x - 4 = 9$, giving $x = 13$, or $x - 4 = -9$, giving $x = -5$.

13 Find the distance between -8 and 15 on the number line, using absolute value.

Distance $= |15 - (-8)| = |23| = 23$.

Word problems involving rates and mixtures

14 A store mixes 4 pounds of candy costing 3 dollars per pound with 6 pounds of candy costing 5 dollars per pound. Find the average price per pound of the mixture.

Total cost: $4(3) + 6(5) = 12 + 30 = 42$ dollars. Total weight: $4 + 6 = 10$ pounds. Average price: $\frac{42}{10} = 4.20$ dollars per pound.

15 A factory's production increased from 240 units per day to 300 units per day. Express this change as a ratio in simplest form.

Ratio of new to old: $\frac{300}{240} = \frac{5}{4}$.

Combining percent, ratio, and rate concepts

16 This question has two parts. A store buys a product for 40 dollars and marks it up 60% to set the retail price. (a) Find the retail price. (b) If the store later offers a 25% discount off the retail price, find the discounted price, and determine whether the store still makes a profit compared to its 40-dollar cost.

(a) Markup amount: $40(0.60) = 24$ dollars. Retail price: $40 + 24 = 64$ dollars. (b) Discount amount: $64(0.25) = 16$ dollars. Discounted price: $64 - 16 = 48$ dollars. Since $48 > 40$, the store still makes a profit of $48 - 40 = 8$ dollars per item.

Order of operations and simplifying numerical expressions

17 Evaluate $6 + 2(9 - 3)^2 \div 4$.

Parentheses first: $9 - 3 = 6$. Exponent: $6^2 = 36$. Then: $6 + 2(36) \div 4 = 6 + 72 \div 4 = 6 + 18 = 24$.

18 Evaluate $\frac{5^2 - 3^2}{2^3 - 2}$.

Numerator: $25 - 9 = 16$. Denominator: $8 - 2 = 6$. Result: $\frac{16}{6} = \frac{8}{3}$.

Fractions, decimals, and conversions

19 Convert $\frac{7}{8}$ to a decimal and to a percent.

$\frac{7}{8} = 0.875 = 87.5\%$.

20 Add $\frac{2}{3} + \frac{3}{4}$, giving your answer as a simplified fraction.

Common denominator 12: $\frac{8}{12} + \frac{9}{12} = \frac{17}{12}$.

Sequences: arithmetic and geometric

21 Find the 10th term of the arithmetic sequence $4, 9, 14, 19, \ldots$

First term $a_1 = 4$, common difference $d = 5$. $a_{10} = a_1 + 9d = 4 + 45 = 49$.

22 Find the 6th term of the geometric sequence $3, 6, 12, 24, \ldots$

First term $a_1 = 3$, common ratio $r = 2$. $a_6 = a_1 \cdot r^5 = 3(32) = 96$.

A multi-step number theory and ratio problem

23 This question has two parts. A recipe for 12 cookies requires a ratio of 3 cups flour to 2 cups sugar to 1 cup butter. (a) Find the amount of each ingredient needed to make 30 cookies. (b) If flour costs 0.60 dollars per cup, sugar costs 0.80 dollars per cup, and butter costs 1.50 dollars per cup, find the total ingredient cost for 30 cookies.

(a) Scale factor: $\frac{30}{12} = 2.5$. Flour: $3 \times 2.5 = 7.5$ cups. Sugar: $2 \times 2.5 = 5$ cups. Butter: $1 \times 2.5 = 2.5$ cups. (b) Total cost: $7.5(0.60) + 5(0.80) + 2.5(1.50) = 4.5 + 4 + 3.75 = 12.25$ dollars.